Definition of Relation

Let $A = \{p, q, r, s, t, u\}$ be a set of students and let $B = \{X, Y, Z, W\}$ be a set of schools. Let us consider the following "relation".

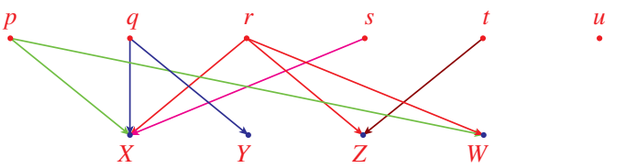
A student $a \in A$ is related to a school $S \in B$ if "a" is studying or studied in the school S.

Let us assume that p studied in X and now studying in W, q studied in X and now studying in Y, r studied in X and W, and now studying in Z, s has been studying in X from the beginning, t studied in Z and now studying in no school, and u never studied in any of these four schools.

Though the relations are given explicitly, it is not possible to give a relation always in this way. So let us try some other representations for expressing the same relation:

(i)
$$\begin{array}{cccccccccc} p & p & q & q & r & r & r & s & t \\ X & W & X & Y & X & Z & W & X & Z \end{array}$$

(ii)



(iii) $\{\ (p, X),\ (p, W),\ (q, X),\ (q, Y),\ (r, X),\ (r, Z),\ (r, W),\ (s, X),\ (t, Z)\ \}$

(iv) $pRX, pRW, qRX, qRY, rRX, rRZ, rRW, sRX, tRZ.$

Among these four representations of the relation, the third one seems to be more convenient and comfortable to deal with a relation in terms of sets.

The set given in the third representation is a subset of the Cartesian product $A \times B$. In Illustrations 1.1 and 1.2 also, we arrived at subsets of a Cartesian product.

Definition 1.2

Let A and B be any two non-empty sets. A ***relation*** R from A to B is defined as a subset of the Cartesian product of A and B. Symbolically $R \subseteq A \times B$.

A relation from A to B is different from a relation from B to A.

The set $\{a \in A : (a, b) \in R \text{ for some } b \in B\}$ is called the ***domain*** of the relation.

The set $\{b \in B : (a, b) \in R \text{ for some } a \in A\}$ is called the ***range*** of the relation.

Thus the domain of the relation R is the set of all first coordinates of the ordered pairs and the range of the relation R is the set of all second coordinates of the ordered pairs.

Illustration 1.3 Consider the diagram in Figure 1.8. Here the alphabets are mapped onto the natural numbers. A simple cipher is to assign a natural number to each alphabet. Here a is represented by 1, b is represented by 2, $\ldots$, z is represented by 26. This correspondence can be written as the set

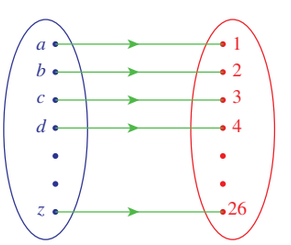


Figure 1.8